Standard Model Gauge Structure under Complete-Response and Observer-Transport Premises

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Abstract

Gauge structure is proposed to be the transport of overlap data between observer patches. On twelve ports with icosahedral incidence, a faithful compact commutator-closed response whose proper rechartings act by inner automorphisms of that same response has exactly one compact Lie type, $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$. Faithfulness fixes the dimension twelve, the single fixed line of the transitive A_5 action bounds the centre by one, the fixed-space count excludes $\mathfrak{su}(2)^4$, and the classification of compact simple Lie algebras, a declared input, leaves the split $11 = 3+8$. The dimension arithmetic and the centre bound are machine checked; the fixed-space exclusion and the classification enter Lean as declared hypotheses.

A conditional matrix realization supplies the current blocks. Its declared spin lift and trace redistribution are the standard $\mathfrak{su}(5)$ embedding, and its declared exterior module is the Georgi–Glashow $\mathbf{10} \oplus \bar{\mathbf{5}}$ generation with exact Standard Model charges, anomaly cancellation and common central kernel $\mathbb{Z}_6$. In the conditional window $3 \leq N_g \leq 5$, whose lower edge is a declared CP-capability clause, a declared band-selection rule and the exact Laplacian costs $5 - \sqrt{5} < 6 < 5 + \sqrt{5}$ select the rank-three band. On the executable carrier the twelve-current law is a specification: the tested native exchange response closes to seventeen dimensions, and no result derives the twelve-dimensional current, the global gauge group or measured couplings from unchanged repair.

Interpretation and state variables. OPH proposes one structure with two descriptions: a timeless, self-referential complete history and its actual internal computation. Observers are proposed to reverse-engineer their reality's substrate and construct hardware whose execution realizes that same history, including its builders and hardware, in emergent, mutually calibrated local times. The candidate must supply construction, initialization, controls, memory, read/write and record services, clock comparisons, and their resource costs; no external host or additional time is assumed. Its realization map must specify whether it matches one actual history, a family of preparation and intervention responses, or every independently allowed microstate. These are different requirements. A candidate must specify the history domain and closure operator; neither the schematic equation nor the finite constructions establish this physical realization or select a unique universe. An actual quantum record history also need not assign simultaneous outcomes to incompatible unperformed measurements.

The proposed primitives are patch states, records, interfaces, and admissible operations. Physical space, duration, and matter are intended emergent readouts. A supplied carrier incidence, geometric domain, action, representation, or clock is a premise of the construction using it. Complete-history closure, an instantaneous repair fixed point, a public normal form, and a stationary probability law are different objects. A stationary law can support changing individual histories. An operational observer is defined by its access and record interfaces; consciousness requires an additional psychophysical interpretation.

Basis, premise labels and paper set. OPH has one hypothesis, closure: the universe is a timeless self-referential structure whose observers recover its specification from their records and realize it. Its internal-realization policy requires its operations and information access to be accounted within the structure; operational agreement on accepted shared data defines what is public. The executable specification of carrier, repair law, observer algebra and closure inputs is implemented and runs; it is a candidate judged by whether it reproduces what observers measure. The labels A1, A2 and A3 name theorem premises: the carrier, federation and response layer of that specification; the agreement condition with an additional response-endogeneity principle; and an adopted least-informative state-selection principle. Closure alone implies neither the entropy minimizer nor inner implementation of rechartings on the connected response algebra. A flagship synthesis gives the account, an executable observer construction states the specification with its acceptance criteria and premise map, and the theorem library carries conditional implications whose antecedents that premise map tracks.

Time conventions. Host runtime, solver counters, native attempt counts, local event-clock readings, event precedence or logical numbering, model coordinates or lapse, modular parameters, and calibrated observer-clock readings have distinct roles. Their identification requires explicit maps. A modular parameter is dimensionless; its geometric and operational readings require the stated branch and clock premises. Consistent local clock comparisons require no preferred universal clock or identical clock rates. Shared time parameters in restricted benchmarks are declared model choices. Experienced succession and a final instantaneous cosmic endpoint do not follow from a modular-flow or finite-convergence theorem.

This paper's hypothesis and place in the tower

This paper is the gauge half of layer 5. Its hypothesis is that gauge structure is the transport of overlap data between observer patches: on the twelve-port icosahedral carrier, a complete faithful commutator-closed port response on which every proper carrier recharting acts endogenously, by an inner automorphism of the same response, is the local gauge current, and its compact Lie type is $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$.

What is proved. The Lie-type theorem (Theorem 3.1) is a finite implication from the response clauses of premise A1, the endogenous-transport clause of premise A2 and the classification of compact simple Lie algebras. The categorical route proves a fixed-cutoff bosonic sector category unconditionally and a compact Tannaka group on any refinement tail that carries the stated receipt; a Doplicher–Roberts field net requires further declared hypotheses. The matrix current, spin lift, trace redistribution, exterior matter module, band selection, bracket discriminator and local action are conditional realizations under displayed declarations.

Open joins. The executable model must realize the complete response of A1 and the endogenous transport of A2 from its own repair and memory law, with the port generators, commutators and

implementers reconstructed from ordered source histories. The tested native exchange response gives seventeen currents and fails A1; a specified coherent control law gives the twelve closed currents on a declared control menu (Section 4). Physical spin and statistics, a physical Weyl field, the global quotient, the family attachment and all couplings are further joins. The matter hypothesis, that stable massive matter is a localized topological defect of native overlap transport whose boundary holonomy takes values in this gauge structure, is developed in the particle paper [1].

Refutation. The complete accessible current of a specified executable law, if it is not a faithful closed twelve-dimensional response with inner carrier action, excludes that law's realization of the branch. A proof that no admissible local repair and memory law on the carrier supplies the A1 and A2 response clauses would remove this gauge branch. The finite failure conditions are listed in Section 12.

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1 Two reconstruction routes

The first route is structural:

$$\text{overlap gluing and transport} \Longrightarrow \text{rigid symmetric sector category} \Longrightarrow \mathrm{Aut}_{\otimes}(\mathcal{F}).$$

It reconstructs a compact group on the declared bosonic branch (Section 5). It does not select the Standard Model without a realized sector witness.

The second route is finite and carrier-specific:

$$\begin{aligned}&\text{twelve-port incidence and complete reversible response}\\ &\Longrightarrow \text{internal proper-carrier holonomy} \Longrightarrow \mathfrak{su}(3)\oplus\mathfrak{su}(2)\oplus\mathfrak{u}(1)\\ &\Longrightarrow \text{conditional Standard Model matter image.}\end{aligned}$$

Incidence supplies the twelve-dimensional A_5-module and its single fixed line. Premise A1 supplies the faithful compact commutator-closed response on that module. Premise A2 makes the proper carrier action inner on the same response. Incidence without the response clauses leaves many equivariant laws. Response naturality without endogeneity leaves the abelian $\mathfrak{u}(1)^{12}$ and the mixed $\mathfrak{su}(2)^2\oplus\mathfrak{u}(1)^6$ branches available; endogeneity bounds the centre by the single fixed line, and the fixed-line count then excludes $\mathfrak{su}(2)^4$. The two routes have separate premise sets. No theorem identifies the finite current with the Tannaka group; that equality requires a same-source current construction and a commuting-square identification.

2 Premises

The premises A1–A3 are shared with the other layers. The Lie-type theorem uses only the response clauses of A1 and the endogenous-transport clause of A2. A3 enters only the edge-sector weighting of Section 10. Branch-local hypotheses are stated at the results that consume them.

A1 (oriented observer-patch federation and spherical support). There exists an observer patch net on an oriented spherical screen: at every finite resolution each local carrier has twelve primitive boundary ports forming the vertices of an oriented triangular boundary with 30 edges and 20 faces, combinatorially the boundary of an icosahedron, and carriers join through typed seams and coherent triple overlaps, refine to an oriented spherical support, and expose local state, readback, records, repair moves, and checkpoints. Their complete infinitesimal port response is represented faithfully on a finite-dimensional unitary response space and is closed under the commutators generated by ordered response composition. Formally, for every regulator r in a directed system there is a typed object $\mathfrak{N}_r = (\mathcal{P}_r, \mathcal{A}_r, \mathcal{R}_r, \mathcal{I}_r, \mathcal{U}_r, \mathcal{C}_r, N_r, S_r, b_r)$: a finite patch/overlap category with an isotone local algebra net and central record algebras; a federation of finite carriers, each with twelve primitive pairwise-orthogonal central port projections summing to one and a boundary packet $K = (P, E, F, o)$ with $|P| = 12$, $|E| = 30$, $|F| = 20$, degree-five ports, five-cycle links, and coherently oriented two-face edges, isomorphic to the icosahedral boundary complex with no preferred labels; seam algebras with unital restrictions and coherent triple-overlap cocycles forming the nerve N_r; a finite edge-midpoint refinement support S_r realized orientation-preservingly in S^2 with mesh tending to zero; and a source-bound degree-one bridge $b_r : N_r \to S_r$, all commuting with refinement. Each carrier also has $V_{r,i} = \mathbb{R}[P_{r,i}]$, a finite-dimensional complex Hilbert space $H_{r,i}$, and an injective real-linear map

$$D_{r,i} : V_{r,i} \longrightarrow \mathfrak{u}(H_{r,i}).$$

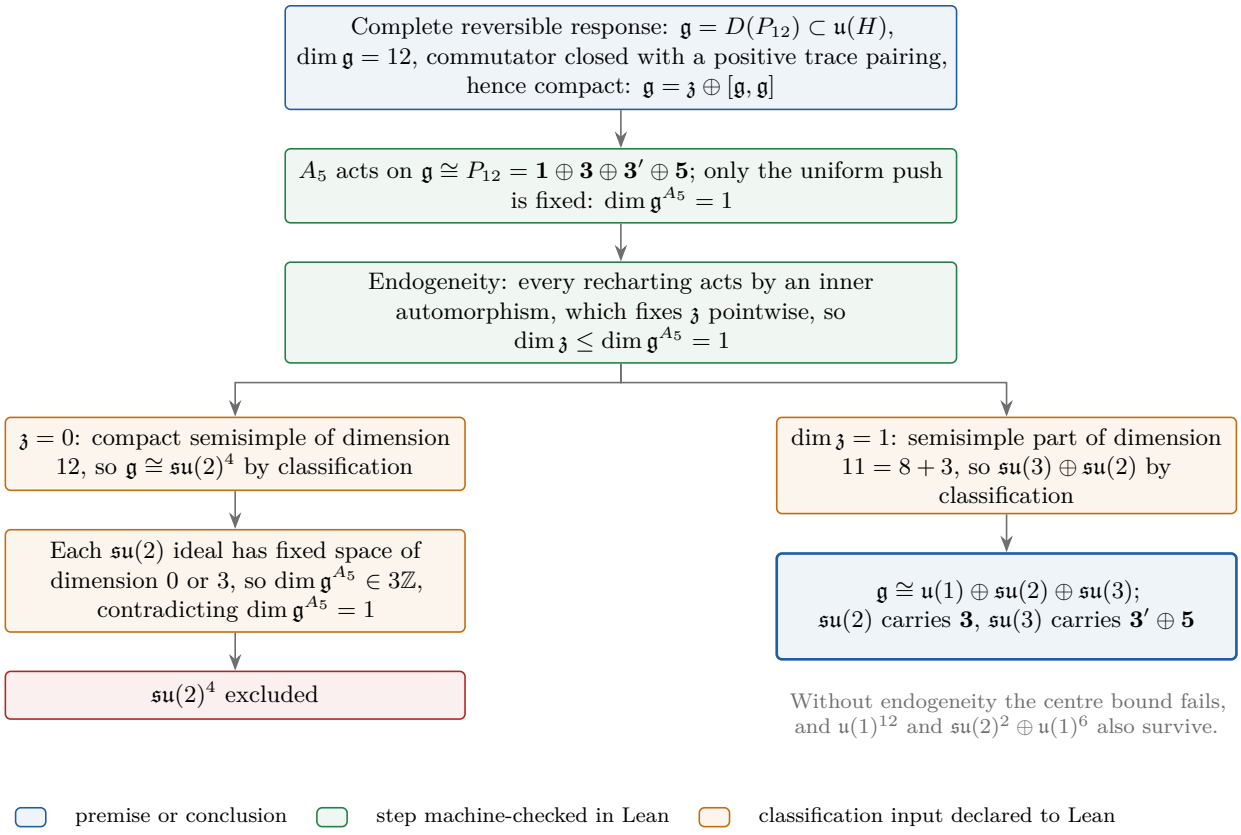


Figure 1: Steps of the gauge Lie-type theorem. Green steps are machine-checked in the Lean library; orange steps use the classification of compact simple Lie algebras and the fixed-space argument against $\mathfrak{su}(2)^4$, which enter Lean as declared hypotheses. The port count fixes the dimension twelve by declaration; the content of the theorem is the exclusion of $\mathfrak{su}(2)^4$ and a one-dimensional centre.

The image $\mathfrak{g}_{r,i} = D_{r,i}(V_{r,i})$ is closed under commutators and complete for the declared quotient-visible infinitesimal port response. The primitive port probes span $V_{r,i}$, no additional public response direction is omitted, and $-\operatorname{Tr}(D_{r,i}(v)D_{r,i}(w))$ is positive definite. The response construction is natural under admissible presentation equivalence and refinement. A1 does not select a particular response table, inverse-port law, Lie type, global group, particle interpretation, or metric content.

A2 (observer agreement). Observers operating on the screen agree on the meaning of the data they jointly interpret. Formally, the interpretation map $\mathcal{J}_r : \mathsf{Data}_r \to \mathsf{Meaning}_r$ from observer-accessible data to operational meanings is natural with respect to every visible overlap restriction, recharting, seam translation, higher-overlap map, federation map, and refinement map: every declared data-access diagram for accepted public data commutes after interpretation, and $\mathcal{J}_r \circ c_{s\to r} = C_{s\to r} \circ \mathcal{J}_s$ across resolutions. A2 constrains accepted shared data only, and the domain it quantifies over is supplied by the A1 interfaces rather than by A2 itself.

Accepted reversible overlap transports form a groupoid $\mathcal{O}_r$. For a complete carrier chart o, its closed paths $\mathrm{Hol}_r(o)$ map onto the orientation-preserving incidence automorphisms $\mathrm{Aut}^+(K_{r,i})$. For every proper carrier automorphism a, A2 supplies a closed path γ_a and one projective implementer

$[U_a] \in \mathrm{PU}(H_{r,i})$ satisfying

$$\mathrm{Ad}_{[U_a]}\, D_{r,i}(v) = D_{r,i}(a \cdot v).$$

Writing

$$G^0_{D,r,i} = \langle \exp(tD_{r,i}(v)) : t \in \mathbb{R},\ v \in V_{r,i} \rangle^0,$$

the implementer is endogenous:

$$[U_a] = [g_a c_a], \qquad g_a \in G^0_{D,r,i}, \quad c_a \in C_{U(H_{r,i})}(\mathfrak{g}_{r,i}).$$

Closed paths compose projectively and the construction commutes with refinement. A unitary lift need not be unique. Scalar phases and centralizer actions are permitted because they act trivially on the response algebra. An independent spectator carrying a matching projective action cannot replace the response-generated factor or contribute a public response direction. The closed path, port action, and implementer must arise from the same accepted source history in an executable realization.

The two response clauses support an abstract target-free compact Lie-type theorem. They do not certify that an executable producer emitted a particular current fixture. Such a certification must reconstruct the port generators, commutator, and projective implementers from ordered source histories without a named current model upstream. A2 does not imply global state extension, repair termination, confluence, unique normal forms, record durability, Byzantine safety, dissemination, or commutation of state optimization with refinement; raw mismatch and repair precede acceptance.

A3 (conditional maximum randomness). Everything that observer agreement leaves unconstrained is maximally random. Formally, at finite regulator r a state is a compatible family of local normalized states on the accessible algebra net; $\mathcal{K}_r$ is the nonempty convex set of such families satisfying the finite observer-visible constraints supplied by A1 and A2 through an A1-generated constraint grammar with a factorization theorem; and, relative to an exact reference family τ_r, an A1-generated observer cover $\mathcal{G}_r$, and strictly positive exact weights $w_{r,P}$ constructed from quotient-visible A1 data by a rule natural under admissible presentation equivalence, the realized state is the information projection. The cover restriction map is injective on $\mathcal{K}_r$; equivalently, at smooth points no nonzero feasible tangent is invisible to the whole cover. The realized state is

$$\rho_r = \arg\min_{\rho \in \mathcal{K}_r} \sum_{P \in \mathcal{G}_r} w_{r,P}\, D(\rho_{r,P} \| \tau_{r,P}).$$

The same projection rule applies separately to each declared optimizer object type (ontic state, inference state, or transition distribution), and statements about one type do not transfer to another. This is equivalent to weighted local entropy maximization when every local reference density is identity-proportional in its declared trace. "Random" means least informative relative to the declared reference and the complete agreement constraint set. A3 selects one state inside one fixed feasible space. It does not compare unrelated state spaces or field contents, does not imply optimizer compatibility across refinement, and does not imply recovery, alignment, or mixing; each such statement carries its own theorem, interface, or countermodel.

The complete formal statement of A1–A3, including its constraint map and non-implications, is given in the premise registry.

3 The finite-current Lie-type theorem

Theorem 3.1 (Finite-current fixed-space theorem)**.** *Let $\mathfrak{g} = D(\mathbb{R}^{12}) \subset \mathfrak{u}(H)$ be the image of the complete reversible twelve-port response of premise A1, and let the proper carrier rotations act on it by inner automorphisms of the same response, as premise A2 requires. Then*

$$\mathfrak{g} \cong \mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3).$$

The implication assumes no ambient continuous gauge group and is independent of the categorical refinement receipt.

Proof. Faithfulness gives $\dim \mathfrak{g} = 12$. Commutator closure inside $\mathfrak{u}(H)$ makes $\mathfrak{g}$ compact reductive, so $\mathfrak{g} = Z(\mathfrak{g}) \oplus [\mathfrak{g}, \mathfrak{g}]$. The transitive action of the alternating group A_5 on the twelve ports gives $\dim \mathfrak{g}^{A_5} = 1$. Inner automorphisms fix the centre pointwise, so $Z(\mathfrak{g}) \subset \mathfrak{g}^{A_5}$ and $\dim Z(\mathfrak{g}) \leq 1$. If the centre vanished, $\mathfrak{g}$ would be semisimple of dimension twelve; the compact simple algebras of dimension at most twelve have dimensions 3, 8 and 10 [2], so $\mathfrak{g} \cong \mathfrak{su}(2)^4$. The inner action preserves each simple factor of $\mathfrak{su}(2)^4$; simplicity of A_5 makes each restricted action trivial or faithful, with fixed dimension three or zero. No sum of such terms equals one, so this branch is excluded. Hence $\dim Z(\mathfrak{g}) = 1$, the semisimple part has dimension eleven, and compact-simple classification gives the unique split $11 = 3 + 8$. □

The port count fixes the dimension twelve by declaration; the content of the theorem is the exclusion of $\mathfrak{su}(2)^4$ and a one-dimensional centre. The dimension arithmetic, the centre bound and the dimension-six exclusion below are machine checked in the accompanying Lean development [3]. The $\mathfrak{su}(2)^4$ fixed-space step and the compact-Lie classification enter Lean as declared hypotheses, and no source current or full Lie-algebra isomorphism is constructed there. The formal interface keeps the inputs separate: an injective port-response map supplies dimension twelve; an inner carrier action places the centre inside the one-dimensional fixed space; the fixed-space exclusion supplies a nonzero centre; and compact-simple classification supplies the allowed simple dimensions $3, 8, 10$. The final Lean selection theorem takes the last three constraints as hypotheses and proves the split. An observable relative phase alone establishes neither centrality nor trivial carrier action on the whole centre. The trivial carrier action on the centre is a sufficient premise for this Lie type; the proof supplies no independent physical origin for it.

What incidence alone supplies. The certified incidence and inverse-port pairing determine J, with

$$10J = A^3 - 4A^2 - 5A + 10I,$$

where A is the port adjacency. The equivariant linear-response commutant is four-dimensional, and a target-blind impulse and readback protocol produces $R = -J$. Its restrictions to $\mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}$ have fixed relative signs; simultaneous reversal is an algebraic response convention and is not identified with matter charge conjugation. These facts constrain the response without constructing the current derivative.

Exact negative controls. The registered single-recurrence interface generates the commutative algebra $\mathrm{span}\{I, A, A^2, A^3\}$. It contains neither twelve independent current generators nor a nonzero bracket, and its only proper recharting is the identity; this bound applies to that registered packet. The target-free diagonal phase lift has twelve independent first derivatives and an abelian bracket. Adding the connected adjacency tangent generates $\mathfrak{u}(12)$, with derived rank 143. The canonical

scalar seam equalizers do not supply the current either. On the eleven-dimensional zero-sum port space,

$$\operatorname{rank}\langle [G_e, G_f] \rangle = 49, \qquad \operatorname{Lie}\langle [G_e, G_f] \rangle = \mathfrak{so}(11), \qquad \operatorname{Alg}\langle G_e \rangle = \operatorname{End}(\mathbb{R}^{11}),$$

of dimensions 49, 55 and 121. The strict-descent repair maps therefore cannot be relabelled as the twelve-dimensional compact reversible current, and the repair law acquires no gauge content by definition.

Carrier specificity. Identify the twelve ports in antipodal pairs. The quotient triangulates the real projective plane with six vertices, fifteen edges and ten triangular faces; its edge graph K_6 has twenty 3-cycles. Its transitive A_5 action retains one fixed line. If a six-dimensional compact current carries that action internally, a one-dimensional centre leaves the impossible semisimple dimension five, whereas a zero-dimensional centre gives $\mathfrak{su}(2)^2$ with fixed dimension zero, three or six. No current satisfies all those premises. The adjacency spectrum $5, -1$, with multiplicities one and five, is the antipode-even part of the icosahedral spectrum; the two $\pm\sqrt{5}$ triplets are antipode-odd. Antipodal identification changes topology and port count; forgetting a chosen orientation alone leaves the twelve-port sphere.

4 The current on the executable carrier

The Lie-type theorem consumes a complete response; it does not produce one. On the executable carrier the twelve-current law is a specification; no result derives it from the unchanged native repair. Fixed unrecorded native seam channels annihilate nonzero operators and cannot supply invertible currents. Active correction has a different domain, examined here on one protected native object [4, 3].

A persistent geometric object. The thirty-six nonedge pairs of the native binary load-two carrier admit two source-derived equivariant perfect-matching colourings by the six antipodal axes. Each normalized matching code has dimension six, carries the actual port-permutation representation $1 \oplus 5$, and corrects every complete recorded C/E branch on all thirty native seams with the declared coherent encoder and record-dependent recovery. Its sum-zero quintet is protected. Two such corrected objects contain a unique geometric $3 \oplus 3'$ sector under simultaneous port rotations, with an exact intertwiner inside the full thirty-six-dimensional product code. This is a persistent geometric carrier; it supplies no spin-half lift, exterior matter module or autonomous selection of that sector.

The native exchange response fails the complete-response clause. On opposite colouring choices, the twelve corresponding-port exchanges on all 10626 native load-four configurations yield six independent Hermitian mixed responses that preserve the geometric sector. They use the declared joint-code phase control and are not copied from any target current. Their exact real Lie closure is

$$\mathfrak{g}_{\text{native}} = \mathfrak{su}(3) \oplus \mathfrak{su}(3) \oplus \mathbb{R}\, iI_6, \qquad \dim \mathfrak{g}_{\text{native}} = 17.$$

The proper port rotations belong to this connected response group at the effective-algebra level, so the response realizes an endogeneity bridge on one object. It does not instantiate A1. Conjugation on $\operatorname{End}(\mathbb{C}^6)$ sees sixteen directions, since the common scalar phase is silent there. On the full

product code the same preserving menu has nine independent Hermitian operators with a three-dimensional restriction kernel, so its Lie closure has dimension at least twenty and its conjugation-visible response on $\mathrm{End}(\mathbb{C}^{36})$ has dimension at least nineteen. The response lacks the relative phase between 3 and $3'$ and allows five traceless phase directions on each triplet. The exchange law and the specified law below intersect in eleven dimensions, and exact quotients and rotation-fixing channels cannot delete the unwanted quintet while supplying the missing centre. This seventeen-versus-twelve mismatch excludes the exchange grammar; it does not exclude other admissible current laws.

A specified twelve-current law. A distinct specified coherent law realizes the current interface on the same protected object without changing its ports or C/E instruments. For the native port vectors v_p, Galois conjugation σ and cross-product matrix $\widehat{v}$, set

$$D_p = \mathrm{diag}\left(\widehat{v_p/2} + iQ_p,\ \widehat{\sigma(v_p)/2}\right), \qquad Q_p = \tfrac{1}{2}v_p v_p^T - \tfrac{1}{6}\|v_p\|^2 I_3 + \tfrac{1}{12}I_3.$$

The native near-fivefold class sum distinguishes the two triplets before the complex-versus-real control choice is applied. Exact checks give rank twelve, all sixty-six bracket closures and all seven hundred twenty port covariance identities. Proper carrier rotations belong to the same connected current group. The sum of the currents is $\mathrm{diag}(iI_3, 0)$, whose relative phase is observable by cross-triplet readout. The algebra is $\mathfrak{u}(3) \oplus \mathfrak{so}(3) \cong \mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$. Source words execute its primitive currents, composed with the geometric preparation, native repair, record-assisted recovery and pointer outputs. Completeness holds over the specified current-control menu. The coherent compiler and detector services are declared apparatus, and excluding them from the menu does not establish their physical inaccessibility, full physical A1 closure or gauge redundancy.

Controlled limits. A positive native class-sum bias, fresh-bath coupling and a complete eighteen-coefficient disturbance-gradient controller give a source-executed bounded-drive joint limit that approaches twelve observable currents. Every finite filter is invertible on all eighteen input currents; fixed-bias long feedback instead leaves six observable currents. Reference cooling alone leaves every work current available. The bias, baths, actuator and feedback rule are architecture choices that unchanged C/E repair does not select.

4.1 Carrier rechartings, the shared triplet and physical rotations

One three-dimensional A_5 representation on the port vectors appears in three roles. The vectors v_p span the adjacency eigenspace with eigenvalue $\sqrt{5}$, the Laplacian band $L = 5I - A$ of cost $5 - \sqrt{5}$. This band is the rank-three Gram frame and slow repair band that supplies the spatial frame triplet [5, 6]; it is the band selected as the family candidate in Section 7; and the colour block of the specified current acts on it, each generator containing the frame rotation $\widehat{v_p/2}$. Its Galois conjugate $\sigma(v_p)$ spans the $-\sqrt{5}$ band $3'$ on which the weak $\mathfrak{so}(3)$ block acts.

The paper keeps these roles distinct. Premise A2 places the finite carrier rechartings inside the connected internal current group, up to centralizer elements and phases; this is a finite covariance statement and it is a premise. No result here identifies those rechartings with spatial rotations of a recovered interacting field, and representation equivalence alone proves neither an identification nor an independence of physical colour, family and spatial actions. Equivalent triplets can occur on different tensor factors: the specified family-feedback construction uses a gauge-trivial ancillary triplet rather than the coloured current triplet. The local rank-three Gram band and the refining

global low band are also distinct: the former has a fixed nonzero carrier energy, whereas the latter concentrates on cell averages.

The conditional spin and matter join constructs commuting actions on $\mathbb{C}^2 \otimes \mathbb{C}^{15}$, with physical-spin central element $-I$ on all thirty states. The diagonal locked action instead has fourteen minus and sixteen plus central eigenvalues and fails that fermionic test. An explicitly chosen compensating internal transport realizes the independent spin action. These are finite existence and obstruction results under a chosen tensor attachment, not a source-selected spacetime connection or a joint charged propagator.

Identifying carrier rechartings with physical rotations while A2 keeps them internal would combine spacetime and internal symmetry. In a relativistic interacting scattering theory that combination is constrained by the Coleman–Mandula theorem under its Poincaré, spectrum, analyticity and nontrivial-scattering hypotheses.[1] The paper makes no such identification, and those hypotheses are not established for the finite interface. The required test identifies rotations operationally on the same clocked field response, establishes their independence from internal currents up to the declared gauge equivalence, including any allowed compensators, and checks that the dynamics preserves that identification.

5 The categorical route

5.1 Transport from overlap gluing

At a fixed regulator the overlap rechartings of a connected cut Σ form a compact unitary transition system. We use chronological multiplication of comparison maps: U_{ij} maps components in chart j to components in chart i, so an overlap path $p = (i_0 \to \cdots \to i_m)$ has $U_p = U_{i_0 i_1} \cdots U_{i_{m-1} i_m}$. Thus U_p compares the terminal chart to the initial chart; forward transport of column components is $U_p^\dagger$. If $p * q$ means first following p, then q, the conventions are $U_{p*q} = U_p U_q$ and $U_{\overline{p}} = U_p^{-1}$ for the reversed path $\overline{p}$. Gauge-as-gluing, the tree and loop lemmas and the transportability theorem are proved in the spacetime paper [6]. The gauge route consumes the following exact criterion.

On the central-defect branch, identify the overlap centres with one fixed abelian unitary coefficient group Z_Σ on which overlap transport acts trivially. With $U_{ji} = U_{ij}^{-1}$, define the implementer-level multiplier

$$\Omega_{ijk} = U_{ij} U_{jk} U_{ki} = z_{ijk} \in Z_\Sigma,$$

equivalently $U_{ij} U_{jk} = z_{ijk} U_{ik}$. The multiplier is defined at the implementer level: $\mathrm{Ad}(z_{ijk})$ is the identity and does not record the defect.

Theorem 5.1 (Central defect strictification and residual holonomy)**.** *The family $\{z_{ijk}\}$ is an untwisted Čech 2-cocycle with coefficients in Z_Σ, and its class $[z] \in \check{H}^2(N_\Sigma, Z_\Sigma)$ is gauge invariant. A central edge rephasing by a cochain in $C^1(N_\Sigma, Z_\Sigma)$ removes every triangle defect if and only if $[z] = 0$. After such a rephasing, let U^{str} be the resulting edge 1-cocycle and $\mathcal{C}_\alpha$ the full transported charge block, including carrier and multiplicity data. Loop-coherent endpoint-only transport exists on that block if and only if, in addition,*

$$\mathrm{Hol}_{\alpha, U^{\mathrm{str}}}([\gamma]) = U^{\mathrm{str}}_\gamma\big|_{\mathcal{C}_\alpha} = \mathbf{1}_{\mathcal{C}_\alpha} \qquad \text{for every } [\gamma] \in \pi_1(|N_\Sigma|, i_*).$$

Proof. Conjugation by Ω_{ijk} fixes every triple-overlap observable, so the multiplier is central there; the fixed-coefficient and trivial-action hypotheses place it in Z_Σ. Comparing $(U_{ij}U_{jk})U_{kl}$ with

[1] S. Coleman and J. Mandula, "All Possible Symmetries of the S Matrix," *Physical Review* **159**, 1251–1256 (1967), https://doi.org/10.1103/PhysRev.159.1251.

$U_{ij}(U_{jk}U_{kl})$ on a quadruple overlap gives $z_{jkl}z_{ikl}^{-1}z_{ijl}z_{ijk}^{-1}=1$. A vertex-frame change $U_{ij}\mapsto V_iU_{ij}V_j^\dagger$ conjugates z_{ijk} and leaves a central multiplier unchanged. A central edge rephasing $U_{ij}\mapsto a_{ij}^{-1}U_{ij}$ replaces z by the corresponding coboundary, so $[z]=0$ is equivalent to removing all triangle multipliers. For two paths p,p' with common endpoints, the residual comparison $U_pU_{p'}^{-1}$ is the holonomy of the closed loop $p*\overline{p'}$, based at their common initial chart. It is the identity for every pair exactly when the displayed holonomy representation is trivial. Conversely, strict path independence makes every triangle comparison and every represented loop action trivial. □

If overlap transport acts nontrivially on a local system of centres, the comparison carries the transported factor and requires the twisted Čech differential. On the genuinely noncentral branch the coefficient data is a crossed module $H_\Sigma\to G_\Sigma$. The vanishing $o_\Sigma^{(2)}=0$ strictifies the higher associator, after which at least one allowed strict G_Σ-valued representative must have trivial represented holonomy. The full orbit q_Σ need not determine a unique ordinary H^1 class and is not itself the strictification test. A nonzero $o_\Sigma^{(2)}$ remains a higher-gauge sector, and an orbit whose every strict representative has residual loop action lies outside the ordinary path-independent sector. The class $[z]$ is structurally analogous to an effective 't Hooft anomaly class; a map between them requires a separate anomaly-descent construction, so $[z]=0$ is used here only as the internal strictifiability condition. The overlap obstruction calculus classifies and routes sectors; it does not select the Standard Model.

5.2 The fixed-cutoff sector category

At cutoff r, choose one common stagewise strict edge cocycle U_r^{str} from the allowed representatives, and retain only seeds on which this representative acts trivially around every closed overlap loop. Sectors requiring incompatible strictifications define alternative fixed-stage categories; they are not unioned before tensor closure.

Definition 5.2 (Fixed-cutoff tensor-generated sector category)**.** *Let S_r be the set of visible collar carriers $W_{\alpha,r}$ with trivial holonomy under the common U_r^{str}. Define*

$$\mathsf{Sect}_r^{\mathrm{bos}}=\langle\mathbf{1},S_r,S_r^*\rangle_{\oplus,\otimes,\text{subobjects,isomorphism}}\subset\mathrm{Rep}_{\mathrm{fd}}(\widehat{K}_{\Sigma,r}),$$

the replete full additive Karoubi rigid tensor subcategory generated by the finitely many visible carriers. Its simple objects are the irreducible summands of finite tensor words in seeds and conjugates; there may be infinitely many of them although every object and Hom space is finite-dimensional.

Theorem 5.3 (Fixed-cutoff bosonic sector category)**.** *At every fixed regulator cutoff, on the ordinary or central-defect zero-obstruction branch and in the bosonic internal-gauge sector of the $3+1$-dimensional effective regime, $\mathsf{Sect}_r^{\mathrm{bos}}$ is a semisimple rigid symmetric C^*-tensor category with one monoidally coherent path-independent transport. The carrier-forgetful functor $F_r:\mathsf{Sect}_r^{\mathrm{bos}}\to\mathsf{Hilb}_{\mathrm{fd}}$ is faithful, *-preserving and symmetric strong monoidal.*

Proof. Complete reducibility of finite-dimensional representations of the compact group gives semisimplicity and subobjects; the operator adjoint and norm give the C^*-structure; conjugate representations give duals; the flip gives the bosonic symmetry. Around a closed loop U_r^{str} acts as the identity on every retained seed, hence on direct sums, tensor products, conjugates and invariant subobjects. Collar concatenation realizes tensor words whenever the finite tensor-realization receipt is supplied. □

Fermionic signs, spinorial matter and chirality are not part of this bosonic category; they belong to the matter lift of Section 6.

5.3 Refinement and compact-group reconstruction

Definition 5.4 (Compact-gauge refinement receipt)**.** *On a cofinal tail, the receipt supplies: a surjectivity certificate for each finite-state restriction $\rho_{sr} : Q_s \to Q_r$; multiplicities $m^{rs}_{\beta\alpha}$ and unitaries V^{rs}_β with $d_{\beta,s} = \sum_\alpha m^{rs}_{\beta\alpha} d_{\alpha,r}$, defining the injective block embedding*

$$\big(\jmath_{rs}(a)\big)_\beta = V^{rs}_\beta \left[\bigoplus_\alpha (a_\alpha \otimes \mathbf{1}_{m^{rs}_{\beta\alpha}})\right] (V^{rs}_\beta)^*$$

of the collar algebras, coherent under composition; continuous surjections $\pi^\Sigma_{sr} : \widehat{K}_{\Sigma,s} \twoheadrightarrow \widehat{K}_{\Sigma,r}$ with $\pi^\Sigma_{tr} = \pi^\Sigma_{sr} \circ \pi^\Sigma_{ts}$; one common strict representative at each stage, intertwined by the pullbacks; and compatible finite concatenated-collar realizations of every tensor word and invariant subobject projector. The finite-state consensus maps alone imply none of these data.

Theorem 5.5 (Refinement functor and fibre descent)**.** *On a tail carrying the receipt, $U_{rs} := (\pi^\Sigma_{sr})^* : \mathsf{Sect}^{\mathrm{bos}}_r \to \mathsf{Sect}^{\mathrm{bos}}_s$ is fully faithful, symmetric strong monoidal and *-preserving; it preserves the unit, duals, irreducibles and the full zero-obstruction criterion, with $U_{st}U_{rs} = U_{rt}$ and $F_s U_{rs} = F_r$ on carrier spaces.*

Proof. Set $U_{rs}(H_X, \varrho_X) = (H_X, \varrho_X \circ \pi^\Sigma_{sr})$ and $U_{rs}(f) = f$. Surjectivity gives

$$\mathrm{Hom}_{\widehat{K}_{\Sigma,s}}(U_{rs}X, U_{rs}Y) = \mathrm{Hom}_{\widehat{K}_{\Sigma,r}}(X, Y),$$

so pullback is fully faithful and a coarse simple cannot split. Pullback commutes with tensor products, unit, conjugates, duality maps and the flip. The receipt intertwines the stagewise strict representatives, so zero holonomy persists for the whole category. Pullback changes only the action, so the fibre comparison is the identity. □

On such a tail, the directed colimit $\mathsf{Sect}_\infty := \varinjlim_r \mathsf{Sect}^{\mathrm{bos}}_r$ retains the sectors and intertwiners that persist, and the stagewise carriers descend to a faithful bosonic fibre functor $\mathcal{F} : \mathsf{Sect}_\infty \to \mathsf{Hilb}_{\mathrm{fd}}$. A realized zero-obstruction class with representatives satisfying $U_{st}\alpha_s \simeq \alpha_t$ determines a unique object of the colimit. Without the receipt only the fixed-stage categories are constructed.

Theorem 5.6 (Receipt-conditional Tannaka reconstruction)**.** *On the ordinary or central zero-obstruction bosonic branch with a cofinal tail carrying the receipt, Sect_∞ is a rigid symmetric C^*-tensor category with faithful bosonic fibre functor $\mathcal{F}$, $G := \mathrm{Aut}_\otimes(\mathcal{F})$ is a compact group, and $\mathsf{Sect}_\infty \simeq \mathrm{Rep}(G)$, with G unique up to isomorphism. If the category is trivial, G is trivial.*

Proof. The refinement functors preserve tensor products, unit, duals and *-structure, so the colimit inherits a rigid symmetric C^*-tensor structure and the carriers descend to $\mathcal{F}$. A unitary symmetric monoidal natural automorphism η embeds as $(\eta_X)_X$ in $\prod_X U(\mathcal{F}(X))$. Naturality, $\mathcal{F}(f)\eta_X = \eta_Y \mathcal{F}(f)$, and monoidality, $\eta_{X\otimes Y} = J_{X,Y}(\eta_X \otimes \eta_Y)J^{-1}_{X,Y}$ with $\eta_{\mathbf{1}} = \mathrm{id}$, are closed relations, so G is a closed subgroup of a compact product. Doplicher–Roberts and Tannaka reconstruction give $\mathsf{Sect}_\infty \simeq \mathrm{Rep}(G)$ [7, 8], and two groups reconstructing the same category with compatible fibre functors are both the automorphism group of that functor. □

The local-Gibbs and collar-mixing package is not used to prove nontriviality or to supply the receipt. The theorem is neutral on whether Sect_∞ is trivial: it returns whatever compact group the tensor-fibre data reconstructs and does not select $(\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1))/\mathbb{Z}_6$.

Definition 5.7 (Field-net realization hypotheses)**.** *(R1) An isotonous net $O \mapsto \mathcal{A}(O)$ of C^*-algebras over the small-region collar localizations, with a faithful vacuum representation. (R2) Spacelike commutativity, Haag duality or its declared essential variant, and Property B. (R3) Each persistent class $[\alpha]$ is realized by a localized endomorphism ρ^α_O. (R4) Transportability by unitary intertwiners in $\mathcal{A}$; vanishing collar holonomy is a necessary input to (R4) and is not equivalent to it. (R5) Finite permutation-symmetric statistics and conjugates matching Sect_∞. (R6) The assignment $[\alpha] \mapsto [\rho^\alpha]$ is an equivalence onto the localized transportable finite-statistics endomorphisms.*

Corollary 5.8 (Two-level reconstruction)**.** *Level (i), proved: on the receipt-certified branch, $\mathsf{Sect}_\infty \simeq \mathrm{Rep}(G)$ with no observable net involved. Level (ii), conditional: under (R1)–(R6) the Doplicher–Roberts theorem [7, 8] supplies a bosonic field net $\mathcal{F}_{\mathrm{net}}$ carrying a G-action whose field operators implement every persistent sector, with $\mathcal{A} = \mathcal{F}_{\mathrm{net}}^G$.*

Proof. Level (i) is Theorem 5.6. Under (R1)–(R6) the localized transportable endomorphisms form a category equivalent to Sect_∞ by (R6), hence to $\mathrm{Rep}(G)$; the Doplicher–Roberts inputs are exactly (R2)–(R6) on the net (R1). □

This paper constructs the categorical side only; no observable net is specified on which the persistent sectors act, so every fixed-point identity $\mathcal{A} = \mathcal{F}_{\mathrm{net}}^G$ is conditional on (R1)–(R6).

Lemma 5.9 (Product factorization)**.** *If $\mathsf{Sect} \simeq \mathsf{Sect}_1 \boxtimes \mathsf{Sect}_2$ identifies the reconstructed fibre functor with $\mathcal{F}_1 \boxtimes \mathcal{F}_2$, where each $\mathcal{F}_i$ is a faithful symmetric tensor fibre functor, then $\mathrm{Aut}^\otimes(\mathcal{F}) \cong G_1 \times G_2$ with $G_i = \mathrm{Aut}^\otimes(\mathcal{F}_i)$ and $\mathsf{Sect}_i \simeq \mathrm{Rep}(G_i)$.*

Proof. A tensor-natural automorphism of the external-product functor restricts to each factor by evaluation on $(X, \mathbf{1})$ and $(\mathbf{1}, Y)$, and the two restrictions determine it on every (X, Y) by tensor naturality. Reconstruction on each factor gives the representation equivalences. □

A bare categorical factorization without compatible fibre functors does not establish the product statement, and factorized heat-kernel weights on one state do not reconstruct a product group. If, independently, the finite Lie-type theorem certifies the two nonabelian factors and the charge theorem certifies one connected U(1), the connected realization is $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ up to the separately computed finite kernel.

6 Conditional matter realization

6.1 The matrix current, spin lift and exterior module

The charged-double-triplet matrices are an exact conditional realization of the premise-implied Lie type. A separate geometric-response construction probes all matrix units of $M_6(\mathbb{C})$ and resolves twelve current generators. The declared spin lift sends a six-state generator $D = \mathrm{diag}(A, \widehat{w})$ to

$$L(D) = \mathrm{diag}\left(A, -\frac{i}{2} w \cdot \sigma - \frac{\mathrm{tr}\, A}{2} I_2\right).$$

Together with the declared central trace redistribution this is the standard embedding $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3) \subset \mathfrak{su}(5)$. The nonvacuum even exterior module $M = \Lambda^2\mathbb{C}^5 \oplus \Lambda^4\mathbb{C}^5$ has dimension fifteen and is the Georgi–Glashow $\mathbf{10} \oplus \bar{\mathbf{5}}$ generation [9]. Inside the declared ten-component exterior menu, the exhaustive 1024-subset anomaly scan selects $P_{\mathrm{even}} - P_{\mathrm{vac}}$ and $P_{\mathrm{odd}} - P_{\mathrm{top}}$ as the unique charge-conjugate pair of rank-15 chiral projectors. Source-computed charges and Casimirs identify Q, u^c, d^c, L, e^c. One declared Higgs doublet gives three invariant interaction lines Qu^cH, Qd^cH^* and

Le^cH^*, and with three family copies these admit three arbitrary complex 3×3 Yukawa matrices. The spin cover, exterior-sector choice and physical fermion field are separate attachments; even auxiliary exterior parity does not establish physical fermion statistics, and the decoder of these labels is a representation check rather than a particle identification [1].

Lemma 6.1 (Factor identification)**.** *Let H be a positive-dimensional compact connected Lie group. (a) If H has a faithful irreducible two-dimensional pseudoreal unitary representation, the connected derived subgroup of the image is* $\mathrm{SU}(2)$ *acting as the doublet. (b) If H has a faithful irreducible three-dimensional representation whose nonabelian factor acts irreducibly and inequivalently to its conjugate, the connected derived subgroup of the image is conjugate to* $\mathrm{SU}(3)$ *on the fundamental, up to finite central kernel. (c) A connected abelian charge factor acting nontrivially on the coupled carrier gives a one-torus in the abelianized group.*

Proof. By Schur the centre acts by scalars, so the derived subgroup acts irreducibly. On $\mathbb{C}^2$ the only connected compact groups acting irreducibly have derived subgroup $\mathrm{SU}(2)$. On $\mathbb{C}^3$, two nontrivial simple factors would force a tensor decomposition of dimension at least four; among compact simple algebras only $\mathfrak{su}(2)$ and $\mathfrak{su}(3)$ have nontrivial irreducibles of dimension at most three, and the three-dimensional representation of $\mathfrak{su}(2)$ is real. Part (c) is immediate. □

Irreducibility in (a) is load-bearing: $\mathrm{U}(1)$ acting by $\mathrm{diag}(z, z^{-1})$ is faithful, pseudoreal and reducible. The intrinsically complex hypothesis in (b) is load-bearing: $\mathrm{U}(1) \times \mathrm{SO}(3)$ acting by zR satisfies every other hypothesis. None of these lemmas enters Theorem 3.1.

Proposition 6.2 (Conditional matter image)**.** *Assume the certified twelve-port carrier, the response and endogenous transport clauses, the declared matrix current, the declared fermionic spin category and the resulting conjugate projector pair. Anomaly-forced determinant balance and tensor descent on every realized matter tensor give: either projector supplies the one-generation chiral module with weak-doublet and colour-triplet roles on a common carrier $\mathbb{C}^3 \otimes \mathbb{C}^2$; the exact Standard Model hypercharge lattice up to simultaneous charge reversal; $N_c = 3$; the conjugation-insensitive kernel $\mathbb{Z}_6$; and compatible scalar charges and Yukawa channels. The maximal faithful matter image is*

$$G_{\mathrm{packet}} = \frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}.$$

The Lie type uses the carrier, the response clauses and classification; the other conclusions use the declared matrix and matter premises. The response-sign convention and the choice between conjugate projectors are separate, and no theorem identifies them. The cover and its $\mathbb{Z}_2$ and $\mathbb{Z}_3$ quotients carry the same local tensors, so local data do not select the physical global form (Section 6.2). The proposition attaches no laboratory current and does not identify G_{packet} with the Tannaka group.

Theorem 6.3 (Hypercharge from anomaly freedom and Yukawas)**.** *Assume* $\mathrm{SU}(N_c) \times \mathrm{SU}(2) \times \mathrm{U}(1)_Y$*, one generation of left-handed Weyl fermions (Q, u^c, d^c, L, e^c), and the Yukawa channels QHu^c, $QH^\dagger d^c$, $LH^\dagger e^c$. Then anomaly freedom and Yukawa invariance fix the hypercharges up to normalization; with $Q = T_3 + Y$ and $Q(\nu_L) = 0$,*

$$Y_Q = \tfrac{1}{2N_c}, \qquad Y_L = -\tfrac{1}{2}, \quad Y_e = 1, \quad Y_u = -\tfrac{N_c+1}{2N_c}, \quad Y_d = \tfrac{N_c-1}{2N_c}, \quad Y_H = \tfrac{1}{2}.$$

For $N_c = 3$: $Y_Q = \frac{1}{6}$, $Y_u = -\frac{2}{3}$, $Y_d = \frac{1}{3}$.

Proof. Yukawa invariance gives $Y_u = -(Y_Q+Y_H)$, $Y_d = -Y_Q+Y_H$, $Y_e = -Y_L+Y_H$. The $\mathrm{SU}(2)^2\mathrm{U}(1)$ condition $N_cY_Q+Y_L = 0$ and the mixed gravitational condition $2N_cY_Q+N_cY_u+N_cY_d+2Y_L+Y_e = 0$ give $Y_L = -N_cY_Q$, $Y_H = N_cY_Q$, $Y_u = -(N_c+1)Y_Q$, $Y_d = (N_c-1)Y_Q$, $Y_e = 2N_cY_Q$; the $\mathrm{SU}(N_c)^2\mathrm{U}(1)$ and cubic anomalies then vanish. The normalization gives $Y_Q = 1/(2N_c)$. Without the Yukawa channels the cubic anomaly leaves two branches exchanging Y_u and Y_d. □

This is the textbook argument applied to the declared packet; the exact rationals reproduce the Standard Model because every discrete input is the Standard Model's. The channel scan selects the compatible scalar charge branch, not scalar attachment, multiplicity or a one-Higgs completion. Per generation there are $N_c + 1 = 4$ weak doublets, so the global SU(2) anomaly [10] is absent on the realized branch; this is a consistency check on $N_c = 3$, not a selector.

Theorem 6.4 (Conditional charge quantization)**.** *If the maximal faithful image is G_{packet} and $Q = T_3 + Y$ on a separately assumed electroweak branch, every colour-singlet representation of the packet has integer charge.*

Proof. The $\mathbb{Z}_6$ quotient identifies $(e^{2\pi i/3}, -1, e^{i\pi/3})$ with the identity. On a colour singlet with weak spin j and $n = 6Y$, this requires $(-1)^{2j}e^{i\pi n/3} = 1$, so Y is an integer for integer j and a half-integer for half-integer j. In both cases $T_3 + Y \in \mathbb{Z}$. □

The statement concerns packet representations, not a quantum field realization. It is consistent with the bound $|q_p + q_e|/e < 1 \times 10^{-21}$ on matter neutrality, with bulk fractional-charge abundance limits near 10^{-22} per nucleon, and with collider exclusions of stable particles with $q \in [e/3, 0.9e]$ up to masses of about 640 GeV at 95% CL; these searches do not discharge its premises.

Corollary 6.5 (No simple-GUT X/Y gauge channel)**.** *On the connected* $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ *realization and the twelve-dimensional current algebra, the adjoint is* $(8,1,0) \oplus (1,3,0) \oplus (1,1,0)$. *There are no gauge generators in* $(3,2,\pm 5/6)$*, so no ordinary X, Y exchange inside the current algebra.*

Heavier modes outside the algebra, scalar mediators and other ultraviolet mechanisms are not excluded. Conditional on the one-generation table, its accidental baryon and lepton labels and the declared local dimension-six grammar, an exhaustive census of the 715 four-field multisets admits four Hermitian-conjugacy classes, $QQQL$, QQu_Re_R, QLu_Rd_R and $u_Ru_Rd_Re_R$, whose exact exterior contractions remain nonzero after the Schouten, Fierz and identical-field Pauli relations. The diagonal $\mathbb{Z}_6$ generator fixes every field, so the gauge structure and the quotient do not prove proton stability; the census supplies no coefficient or lifetime. In the same way the mixed hypercharge anomaly gives $k_R^{YWW} = N_g(3Y_Q + Y_L) = 0$, so the $\mathbb{Z}_6$ deck phase supplies no direct $\Theta_R W\widetilde{W}$ baryogenesis coupling; a global $B + L$ attachment would give $k_R = 2N_g$.

6.2 Global form

Four central global forms act consistently on the same local tensors: the cover and its $\mathbb{Z}_2$, $\mathbb{Z}_3$ and $\mathbb{Z}_6$ quotients. The kernel theorem identifies the maximal faithful local matter image. On the finite weight lattice of hypercharge, triality and duality labels in the $q = 6Y$ normalization, a label is fixed by every element of the diagonal $\mathbb{Z}_6$ kernel exactly when

$$2t + 3d + q \equiv 0 \pmod 6,$$

and the kernel acts on any label by multiples of this centre character. Every row of the matter table satisfies the congruence, including both carrier blocks and the scalar; a fractionally charged

singlet is blocked by the generator; and exactly one sixth of the thirty-six lattice labels descend, each character value being carried by six labels. A machine-checked receipt composes this onto the selected matter mask, so every selected component descends through the full kernel. The six-axis artifact computes an order-six flux menu after its coefficient relations are declared; the menu matches the $\mathbb{Z}_6$ quotient and carries the declared matter single-valuedly.

The order-six seam classification keeps two actions separate. Every pure-hypercharge character is enumerated: the fixed-space dimension menus are $\{15, 7\}$ at order two, $\{15, 3\}$ at order three and $\{15, 7, 3, 1\}$ at order six, while the canonical diagonal $\mathbb{Z}_6$ kernel acts trivially on all fifteen matter states. These are different representations of the same abstract cyclic order, and refinement preserves the classified actions. The calculation selects no physical seam character, flux action or 2-representation and does not exhaust every seam grammar. Selection of one physical quotient among the four requires a source-derived complete character category, a same-source loop-to-kernel identity, four-dimensional instanton normalization and theta periodicity, monopole dynamics and a laboratory flux attachment; the congruence derives no physical global form.

Corollary 6.6 (Structural electroweak package)**.** *On a separately declared one-Higgs branch, the electroweak factor has Lie algebra* $\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)_Y$*. A nonzero neutral vacuum vector* $\phi_0 = \frac{v}{\sqrt{2}}\binom{0}{1}$ *of* $H = (1,2)_{1/2}$ *has* $Q\phi_0 = (T_3 + Y)\phi_0 = 0$ *and stabilizer* $\mathrm{U}(1)_Q$*; the package contains* $W^{\pm} = (W^1 \mp iW^2)/\sqrt{2}$*, Z, A and the exact charge operator* $Q = T_3 + Y$*.*

Since $e^{i\alpha T_3} e^{i\beta Y}\phi_0 = e^{i(\beta-\alpha)/2}\phi_0$, vector invariance ties $\beta = \alpha$, while the projective ray $[\phi_0]$ has the two-torus stabilizer $\mathrm{U}(1)_{T_3} \times \mathrm{U}(1)_Y$; projectivization alone does not select the electromagnetic diagonal. A machine-checked composition on the committed doublet carrier proves, for declared $\lambda, v, g, g' > 0$ and Yukawa coefficient y, the minimizer shell $|h|^2 = v^2$, one unbroken line with three Goldstone directions, broken mass values $g^2v^2/4$ and $(g^2 + g'^2)v^2/4$, and a Yukawa bilinear of coefficient yv. The mixing angle, v and the W/Z masses belong to the running and matching surface. On the unbroken factor, a supplied Lorentzian Maxwell action $-\frac{1}{2g_Q^2}\int F_Q \wedge *F_Q + \int A_Q \wedge *J_Q$ with $F_Q = dA_Q$ gives $dF_Q = 0$ and $d * F_Q = g_Q^2 * J_Q$; the reconstruction does not supply that action or its coefficient. In a tensor-network realization a faithful representation of $\mathrm{SU}(3) \times \mathrm{SU}(2)$ needs edge dimension $\chi \geq 5$, and $\chi \geq 6$ only under the additional declared requirement of a single irreducible tensor-product carrier.

7 Families and the generation window

Anomaly cancellation is generation by generation and does not fix the number of generations.

Proposition 7.1 (Conditional generation window)**.** *On a separately declared one-Higgs quark branch over the matter image of Proposition 6.2, with* $N_c = 3$ *and the declared clauses of intrinsic CKM CP capability and weak-sector ultraviolet completability, the admissible window is* $3 \leq N_g \leq 5$*. No count inside the window is selected.*

Proof. An $N_g \times N_g$ CKM matrix has $(N_g - 1)(N_g - 2)/2$ physical CP phases, which vanish for $N_g \leq 2$; the declared CP-capability clause therefore gives $N_g \geq 3$. The one-loop coefficient $b_{\mathrm{SU}(2)} = \frac{22}{3} - \frac{1}{3}N_g(N_c + 1) - \frac{1}{6}$, with the last term from one complex Higgs doublet, is positive exactly when $N_g(N_c + 1) < 43/2$; with $N_c = 3$ this gives $N_g \leq 5$. □

The lower edge is the declared CP-capability clause; the window is the complete numerical content of the two clauses and is not forced by anomaly cancellation, the A_5 graph or the target-free source reduct.

Proposition 7.2 (Conditional screen-band selection)**.** *The twelve-port coefficient module decomposes into bands of ranks* $1, 3, 3', 5$*, with exact operational costs of the nontrivial bands*

$$5 - \sqrt{5} < 6 < 5 + \sqrt{5}.$$

Assume that one complete faithful band in the declared rank window $3 \le r \le 5$ *realizes the family carrier and that the lower cost selects between admissible complete bands. Then the rank-three band of cost* $5 - \sqrt{5}$ *is unique. Tensoring it with the rank-*15 *generation gives a conditional rank-*45 *candidate whose chirality and diagonal* $\mathbb{Z}_6$ *action come from the generation table.*

These implications are kernel-decided in the Lean screen certificate. A response receipt using the graph Laplacian as a unitary generator recovers the same cost bands and places the rank-three residue at the lowest positive generator frequency; the channel conserves mode norms, so this is a frequency statement rather than a relaxation statement. For the four-entry band table, faithfulness excludes the dimension-one band and leaves dimensions $3, 3, 5$; removing the window while keeping faithfulness changes no admissible candidate, so the window does not itself explain triplication. The load-bearing identification is that the family object is a single complete faithful band, compared by the declared cost rule and realized in the matter response. A specified source-executed cooling law computes this rank-three band on the native carrier; selecting that rule and identifying its levels with physical families are inputs.

Generation names require no universally conserved generation charge. Within a sector whose charges and statistics are established, ordering nondegenerate masses is a naming convention, and a physical comparison must retain the eigenvectors, residues and charged-current matrix of the same response. The mismatch of the two left quark diagonalizers determines CKM mixing; a generation label conserved by every charged weak transition would exclude that mixing.[2]

Corollary 7.3 (Declared minimal completion)**.** *Assume Proposition 6.2, the receipt-certified bosonic branch, a one-Higgs scalar attachment, the CP and weak-sector clauses, and a declared physical attachment of the rank-three band to the matter packet. Then the completed packet has maximal faithful image data* $(G_{\mathrm{packet}}, \mathcal{R}_{\mathrm{SM}}, H, \mathcal{Y}_{\mathrm{SM}})$ *with* $N_c = 3$*,* $N_g = 3$ *and*

$$Q_i = (3,2)_{1/6}, \quad u_i^c = (\bar{3},1)_{-2/3}, \quad d_i^c = (\bar{3},1)_{1/3}, \quad L_i = (1,2)_{-1/2}, \quad e_i^c = (1,1)_1, \quad H = (1,2)_{1/2},$$

$i = 1, 2, 3.$

The Lie type uses no clause of the completion; the matter type, charges and $N_c = 3$ are exact inside the declared fixture; $N_g = 3$ and the one-Higgs scalar enter through the completion declaration. Under the fixed-cutoff heat-kernel law of Section 10 every finite-dimensional witness label has positive weight; that positivity concerns bosonic sector labels and supplies no light chiral multiplet. Physical identification of the rank-45 candidate requires a source-derived matter-pole map, continuum spin and locality, physical seam selection, residues, excluded-band and refinement receipts; the absence of other light sectors is unproved.

8 The equivariant bracket variety and a face discriminator

Conditional on the canonical oriented carrier, the target-free space $\mathrm{Hom}_{A_5}(\Lambda^2\mathbb{Q}^{12}, \mathbb{Q}^{12})$ has dimension fourteen and an exact rational Reynolds basis. It is a search space; no bracket is selected.

[2] A. Ceccucci, Z. Ligeti and Y. Sakai, "CKM Quark-Mixing Matrix," in the Particle Data Group review (2024), Section 12.1, https://pdg.lbl.gov/2024/reviews/rpp2024-rev-ckm-matrix.pdf.

Substitution of the general bracket into Jacobi gives 2,640 coordinate equations in 44 target orbits with coefficient-row rank 38. Over $\mathbb{Q}(\sqrt{5})$ that rowspace splits into eleven fixed-line derivation-weight products and a rank-27 residual system that applies on every weight flat. A three-parameter derivation-only Jacobi family supplies no compactness or source selection. Conditional on three named textbook lemmas, the compact real locus of the fourteen-parameter equivariant variety is classified exactly over $\mathbb{Q}(\sqrt{5})$ into three certified bracket families F, G, P. For each family the co-root lattices, integral carrier weights and carrier-admitted global forms are computed; the selected forms are adjoint with zero abelian charge lattice, so the abelian sector is undetermined at carrier level. Noncompact and residual components outside the compact locus are unclassified.

Applying the declared equal-weight cyclic rule to the twenty oriented faces gives $B_{\text{face}} = 60R_{13}$. This bracket is A_5-equivariant and fails Jacobi in exactly 240 of the 2640 coordinates, 120 at $+1$ and 120 at -1. Under the added Euclidean metric on upper-triangular structure constants, orthogonal projection onto the compact locus has the unique nearest family G:

$$d_G^2 = \frac{615 - 123\sqrt{5}}{22}, \qquad d_F^2 = \frac{615 + 123\sqrt{5}}{22}, \qquad d_P^2 = 45.$$

Both F and G projections lie in their open compact strata. On the same 792-coordinate convention, exact primal–dual certificates give

$$d_1(G) = 30(\sqrt{5} - 1) < \inf d_1(F) = d_1(P) = 60,$$

$$d_\infty(G) = \frac{5 - \sqrt{5}}{10} < d_\infty(F) = \frac{\sqrt{5}}{5} < d_\infty(P) = \frac{1}{2}.$$

The ℓ_1 value for F is an infimum approached under scaling, not an attained minimum. The Python replay proves the primal–dual optimization and the Lean theorem checks the radical values and their order. The family verdict holds across the ℓ_1, ℓ_2 and ℓ_∞ scores on the classified compact locus once the coordinate norm is supplied.

The quadratic verdict is robust over the invariant class. The carrier splits multiplicity-free as $1 + 3 + 3' + 5$, the commutant of the port action is spanned by the four spectral projectors, and every invariant carrier inner product is $\alpha P_1 + \beta P_3 + \gamma P_{3'} + \delta P_5$. The induced distances are

$$d_P^2 = \frac{15 - 3\sqrt{5}}{\beta} + \frac{15 + 3\sqrt{5}}{\gamma} + \frac{15 - 3\sqrt{5}}{2}\,\frac{\beta}{\delta^2} + \frac{15 + 3\sqrt{5}}{2}\,\frac{\gamma}{\delta^2},$$

$$d_F^2 = \frac{15 + 3\sqrt{5}}{2}\,\frac{\gamma}{\delta^2} + \frac{15 + 3\sqrt{5}}{\gamma} + \frac{60 + 12\sqrt{5}}{11\beta}, \qquad d_G^2 = \frac{15 - 3\sqrt{5}}{2}\,\frac{\beta}{\delta^2} + \frac{15 - 3\sqrt{5}}{\beta} + \frac{60 - 12\sqrt{5}}{11\gamma},$$

with α absent. Both gaps $d_P^2 - d_G^2$ and $d_P^2 - d_F^2$ are coefficient-positive Laurent polynomials, so P is excluded for every invariant metric. At $\gamma = \beta$ the gap $d_F^2 - d_G^2 = 3\sqrt{5}\,\beta/\delta^2 + (90\sqrt{5}/11)/\beta$ is positive, and for $\beta/\delta \in [1/50, 6]$ the family G is uniquely nearest for all γ, δ. The mirror family F is nearer on the other side of the exact three-scale tie surface, with witness $(\beta, \gamma, \delta) = (8, 1, 1)$. Conjugating $\sqrt{5}$ and swapping $\beta \leftrightarrow \gamma$ maps d_G to d_F. Reversing every face orientation negates B_{face} and leaves the distances unchanged. A channel-diagonal invariant metric that is not carrier-induced reverses the balanced-point selection, so the carrier-induced restriction is load-bearing. On the pinned incidence table, equal normalized face weights and the declared barycentric one-third share give port-dual weight $1/12$ at every port; under the declared diagonal measure-to-metric rule this uniform measure induces the scalar carrier inner product at $(1/12, 1/12, 1/12, 1/12)$, where G is uniquely nearest with $d_G^2 = (3690 - 738\sqrt{5})/11$, $d_F^2 = (3690 + 738\sqrt{5})/11$ and $d_P^2 = 540$.

The equal-weight rule, the coordinate norms, the nearest-point law and the measure-to-metric rule are declared inputs, not consequences of A1–A3, so the verdict is a conditional discriminator

obtained without reference to a measured value. A composed machine-checked surface bundles these premises with the classification inputs and the exterior selection, and proves that the premise-implied type has a unique type-matched nearest family at the balanced point and that the fifteen-state count, the anomaly forms and the $\mathbb{Z}_6$ stabilizer are carried by the bundled selection mask. Among the three family labels and the declared 2^{10} exterior table, the admissible pairs are G with either of two conjugate masks. This does not count points of the continuous G-family or classify physical Standard Model structures.

Over the certified brackets, ad-invariance fixes the kinetic shape block-diagonally with zero cross terms. On carrier-projector forms, families F and G reduce the three-dimensional invariant space to two dimensions, $F:\ w_{3+} = \sqrt{5}\, w_5$ and $G:\ w_{3-} = \sqrt{5}\, w_5$, with P the two-to-two control; simultaneous invariance for F and G leaves the ray $(w_{3+}, w_{3-}, w_5) = t(\sqrt{5}, \sqrt{5}, 1)$, an extra premise. The two-factor history theorem constructs a Gibbs kernel from $aq_1 + bq_2$ with exact factorization; equality of two such kernels identifies λa and λb, so only common scaling survives and a/b is not multiplier gauge. No theorem supplies relative factor coefficients or coupling scales [11, 3].

8.1 An assembled structural action

On one typed premise bundle, the gauge kinetic term (one declared magnitude per invariant carrier sector $3^+ + 3^- + 5$, weighted in the two-parameter G-invariant normal form), the quartic potential $\lambda(|h|^2 - v^2)^2$, the gauge mass form at the chosen minimum and the realified Yukawa line with coefficient y sum to one real function of one field tuple. Exact restriction theorems recover each sector on its slice; the gauge coefficients satisfy the G ad-invariance row and collapse to the mirror ray under the extra F row; the cross-sector generators realizing those rows have coefficient ratio $(1, -\sqrt{5})$, and the control weight $(1, 1, 1)$ breaks invariance with exact value $1 - \sqrt{5}$. A fourth summand carries one complex amplitude per row of the ten-component table; its bilinear $\overline{\psi_i}\psi_j$ is phase-invariant on the selected rows exactly when the charges $q = 6Y$ balance, and the charge column forces the pairing to be diagonal. On the shared doublet and singlet rows the extended action evaluates to the mass bilinear with coefficient yv. Every coefficient is a declared parameter; the construction supplies no source-produced spacetime action, measured coupling, family attachment, global-form selection, chirality or spin attachment. The local covariant action, its Cartan reduction and the fermion current are given in the appendices.

9 Refinement stability and relevant transverse perturbations

The state-side refinement stability used by the gauge branch is a separate optimizer-pushforward and refinement-tail interface: it keeps the finite local constraint family across cutoffs, so that a refinement channel acts on the realized Gibbs family $\omega_\ell(\lambda) \propto \exp(-\sum_{x,a} \lambda_a O_a(x) - \sum_b \mu_b Q_b)$ by an induced map $\omega_L(R_{\ell\to L}(\lambda))$. Coarse-graining a finite-range Gibbs family generically leaves any fixed finite list, so this is a substantive premise; premise A3 supplies the fixed-stage optimizer and not this cross-scale statement or the refinement receipt. The square defects that control reconciliation under coarse-graining are treated in the consensus paper [12].

Proposition 9.1 (Conditional growth of a relevant transverse component)**.** *Assume the local finite-constraint branch of A3, the refinement-tail interface, and a differentiable extension of the refinement maps to an enlarged coupling space with transverse coordinate $g_{\mathcal{O}}$ for a gauge-invariant Lorentz-scalar deformation $\mathcal{O}$. If $\delta g_{n+1} = \Lambda_n \delta g_n$ with $|\Lambda_n| \geq q > 1$, every nonzero infinitesimal transverse component grows by at least q^m after m steps. If the zero-coupling surface is invariant and the trajectory starts on it, relevance alone does not generate $\mathcal{O}$.*

Proof. Iteration gives $|\delta g_{n+m}| = \prod_j |\Lambda_{n+j}||\delta g_n| \geq q^m|\delta g_n|$. Generation from zero concerns the transverse part of the full extended map, which is not a coordinate of $R_{\ell\to L}$ and is not supplied by A3. Exact tuning, an invariant manifold, vanishing operator mixing, cancellations, an additional symmetry or fixed-point dynamics can keep the component zero, so the statement is not an exhaustive protection theorem. □

A gauge-invariant Dirac mass is a relevant scalar when the representations allow it. With nonzero mixing into that coordinate, a nonzero mass perturbation is linearly amplified unless a protecting mechanism intervenes. Chiral matter forbids the Dirac mass by symmetry; vector-like branches with vanishing mixing, tuning or fixed-point protection are not excluded, so the diagnostic does not select chirality by itself.

10 Edge-sector weights and coupling extraction

The edge-centre completion yields sector probabilities p_R on collar boundaries. A conditional heat-kernel weighting relates them to a gauge coupling.

Theorem 10.1 (Fixed-cutoff edge-sector law)**.** *Under premises A1–A3, the fixed-cutoff overlap-gauge realization, a quasi-local local-Gibbs generator on the faithful interior branch, and hypotheses (EH-1) and (EH-2), the regulated same-overlap sector distribution is*

$$p_R = \frac{d_R e^{-t\lambda_R}}{\sum_{R'} d_{R'} e^{-t\lambda_{R'}}},$$

with λ_R the Laplacian eigenvalue on the R-isotypic component and t the collar Gibbs parameter.

(EH-1) The edge generator is the factorwise group Laplacian, $H_{\text{edge}} = \Delta_G$, or a positive combination $\sum_i c_i \Delta_{G_i}$ for a product group; the separate Gibbs parameter is t. For finite groups the declared Cayley generating multiset is conjugacy-invariant and inverse-symmetric, so $\lambda_R = |S| - d_R^{-1} \sum_{s\in S} \chi_R(s)$. Without centrality the Fourier transform is a matrix on each irreducible block and the character average is only its normalized trace, not a sector eigenvalue. This is a declared input: gauge invariance and the Gibbs form force only $H_{\text{edge}} = \sum_R h_R P_R$, and any function of the sector label is gauge invariant and realizable by a local Hamiltonian at fixed cutoff. Within second-order bi-invariant operators the Laplacian is unique up to scale, and one declared Markov chain with a Casimir acceptance rule derives (EH-1); its universality is unproved. (EH-2) The probability space is the one-sided electric-centre edge algebra $\bigoplus_R B(V_R)$, on which the sector projector has rank d_R. On the two-sided space $L^2(G)$ the projector has rank d_R^2, and a two-sided Gibbs trace gives weights $d_R^2 e^{-t\lambda_R}$; the d_R law is a statement about the one-sided algebra.

Proof. The edge degrees of freedom at Σ carry a G-action, modelled for the refinement lift by $L^2(G)$ with its Peter–Weyl decomposition [13]. By (EH-2) the physical one-sided observables are $\bigoplus_R B(V_R)$; by (EH-1) the generator is $\sum_R \lambda_R P_R$. Maximum entropy selects $\rho = Z^{-1} \sum_R e^{-t\lambda_R} P_R$, and $\operatorname{Tr} P_R = d_R$ gives the stated law. □

For $\mathbb{Z}_n$ the Laplacian eigenvalues are $\lambda_q = 4\sin^2(\pi q/n)$. The heat-kernel ansatz assigns the sector probabilities $p_q \propto e^{-t\lambda_q}$, so $t = -\log(p_q/p_0)/\lambda_q$. For SU($N$) the conditional ansatz assigns the probabilities $p_j \propto d_j e^{-tC_2(j)}$, so $t = (\log(p_0/d_0) - \log(p_j/d_j))/C_2(j)$ and $g_{\text{ent}}^2 = t/(2\pi)$. In two-dimensional Yang–Mills the Hamiltonian is the group Laplacian, $H = -\frac{g^2}{2}\Delta_G$ with $E_R = \frac{g^2}{2}C_2(R)$, and a cylinder of area A gives $t = g^2 A/2$ exactly; gluing the open-edge weights gives the closed partition sum $\sum_R d_R^2 e^{-tC_2(R)} = K_t(1)$. The Casimir log-gaps $\Delta_R = \ln(p_0/d_0) - \ln(p_R/d_R) = tC_2(R)$

have parameter-free ratios: for SU(2), $\Delta_1/\Delta_{1/2} = 8/3$, $\Delta_{3/2}/\Delta_{1/2} = 5$; for SU(3), $\Delta_8/\Delta_3 = 9/4$, $\Delta_6/\Delta_3 = 5/2$, $\Delta_{10}/\Delta_3 = 9/2$, $\Delta_{15}/\Delta_3 = 4$, $\Delta_{27}/\Delta_3 = 6$. These are algebraic consequences of the weighting, not source-selected coupling predictions. The repair-gap identification on the compact-gauge branch is the subject of the Yang–Mills supplement [14].

A finite obstruction to universal Casimir weights. Gauge symmetry alone does not imply the heat-kernel law. For a single S_3 plaquette, Gauss reduction gives the exact three-state Hamiltonian

$$H(h) = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 24h & -1 \\ -1 & -1 & 12h-1 \end{pmatrix}, \qquad h \geq 0.$$

Writing its ground energy as $-x < 0$ and $z = x + 12h$, the secular equation is $x + 12h - 1 - 1/x - 1/(24h + x) = 0$. The corresponding sector probabilities obey

$$\frac{p_{\mathrm{sign}}}{p_{\mathrm{std}}^2/(4p_{\mathrm{triv}})} = \left(\frac{z-1}{z}\right)^2 < 1.$$

Thus fitting the standard-sector Casimir weight does not reproduce the sign sector at any finite coupling. The positive-time fit exists only for $h > (1 + \sqrt{3} - \sqrt{2})/24$. This exact counterexample retains the need for (EH-1) and (EH-2); numerical agreement in another phase does not remove them.

11 Classical carrier modes and the quantum-particle criterion

A compact group identifies a symmetry type and a classical field equation a classical branch; neither constructs a propagating quantum particle.

Definition 11.1 (Carrier-mode and quantum-particle receipts)**.** *Fix the invariant conversion speed $c_\star$; its value* $299\,792\,458\ \mathrm{m\,s^{-1}}$ *is exact by the SI definition of the metre. A classical massless carrier-mode receipt for q_X consists of (C1) a stated background and phase, (C2) a quadratic action with positive kinetic coefficient, (C3) a gauge fixing or constraint reduction with physical projector Π_X, and (C4) a positive reduced Hamiltonian, with physical Hessian $K_X = Z_X(\omega^2 - c_\star^2|\mathbf{k}|^2)\Pi_X$, $Z_X > 0$. A quantum-particle receipt additionally requires (Q1) a positive-energy quantization with physical Hilbert space, (Q2) a two-point function whose Källén–Lehmann measure contains $Z_X^{\mathrm{pole}}\delta(\mu^2)$ with $Z_X^{\mathrm{pole}} > 0$, and (Q3) the asymptotic-state or LSZ hypotheses of the phase, including a deconfined sector for a coloured carrier.*

Theorem 11.2 (Action-level modes)**.** *(i) An unbroken $\mathrm{U}(1)_Q$ connection with the Lorentzian Maxwell action $S_Q = -\frac{1}{4g_Q^2}\int F_{Q,\mu\nu}F_Q^{\mu\nu}$, $0 < g_Q^2 < \infty$, and no mass operator, has in radiation gauge $K_{Q,T} = g_Q^{-2}(\omega^2 - c_\star^2|\mathbf{k}|^2)\Pi_T$ with $\operatorname{rank}\Pi_T = 2$. (ii) A compact G with the two-derivative pure Yang–Mills action about the trivial background in a perturbative or deconfined phase has $2\dim G$ transverse modes with $K_{\mathrm{YM},T}^{ab} = \delta^{ab}g^{-2}(\omega^2 - c_\star^2|\mathbf{k}|^2)\Pi_T$ and no hard mass at quadratic order.*

Proof. Gauss' law removes the longitudinal pair, leaving two transverse polarizations with positive Hamiltonian and dispersion $\omega^2 = c_\star^2|\mathbf{k}|^2$. The Yang–Mills Hessian is one Maxwell Hessian per generator; interactions begin at cubic order. These calculations establish (C1)–(C4) and none of (Q1)–(Q3). □

Calling these modes photons or gluons requires the quantum-particle receipt. The perturbative mode does not produce a gauge-invariant asymptotic gluon, and in a confining phase the coloured free pole need not occur in the physical spectrum. The analogous transverse-traceless calculation for a supplied Einstein–Hilbert action gives two classical spin-two modes and no graviton state. Vanishing quadratic mass parameters are action-level statements; Higgs or Stueckelberg completions, medium self-energies, confinement and additional massive fields require their own field-content and phase hypotheses.

12 Failure conditions

The finite chain is exact on its stated premises, and each step has a registered negative control. The Lie-type theorem fails if a faithful compact commutator-closed twelve-port response with inner carrier action is exhibited outside $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$; the conditional matter results fail if the incidence identity, the target-blind response, the equivariant lift, the anomaly scan, the tensor-character enumeration or the refinement persistence receipt fails its control. The categorical route applies only inside its combined strictification, trivial-holonomy, localization and refinement and fibre receipts.

On the executable model the branch is tested by computing the complete accessible current of a specified law, with every admitted control and apparatus service in the menu. A complete accessible response that is not faithful, closed and twelve-dimensional with inner carrier action excludes that law's realization of A1 and A2, as the seventeen-dimensional exchange response excludes its grammar. A physical attachment that forced the carrier rechartings to be spatial rotations of an interacting relativistic field while they remain internal would be constrained by the Coleman–Mandula theorem. The particle paper [1] tests whether localized defects of the same transport carry holonomy valued in this structure.

A Local action, Cartan reduction and fermion current

The appendices carry the local covariant action on the declared matter grammar and the finite current constructions built on it.

A.1 A local gauge-covariant action on the declared matter grammar

Fix an oriented flat Lorentzian chart with metric $\eta = \mathrm{diag}(1, -1, -1, -1)$, its two-component Spin convention, and smooth compactly supported variations. These geometric and Spin data are inputs. Take the declared algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ on its fundamental, dual and trivial carriers. The exterior grammar supplies

$$Q : (3,2)_{1/6}, \quad u^c : (\bar{3},1)_{-2/3}, \quad d^c : (\bar{3},1)_{1/3}, \quad L : (1,2)_{-1/2}, \quad e^c : (1,1)_1.$$

The Higgs choice $H : (1,2)_{1/2}$, family multiplicity, minimal first-derivative kinetic law, scalar polynomial truncation and Yukawa intertwiners are declared. No coupling, mass or Yukawa entry is selected.

Use anti-Hermitian connections, with coupling normalization absorbed into their coordinates, and set

$$D_\mu = \partial_\mu + \rho(A_\mu), \qquad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu]. \tag{1}$$

Fermion values and first derivatives generate a complex exterior algebra. Conjugation is the antilinear anti-automorphism exchanging fermions with independent conjugate generators and reversing

products. Scalar and connection coefficients are even. Write $\bar{\sigma}^\mu = (1, -\sigma^1, -\sigma^2, -\sigma^3)$ and $\epsilon_{01} = 1$. With color index i, weak indices a, b, spin indices α, β and family indices r, s, define

$$T^u_{rs} = \epsilon_{\alpha\beta}\epsilon_{ab} Q^{ia\alpha}_r H^b (u^c)^\beta_{s,i}, \qquad T^d_{rs} = \epsilon_{\alpha\beta} Q^{ia\alpha}_r \overline{H}_a (d^c)^\beta_{s,i}, \qquad T^e_{rs} = \epsilon_{\alpha\beta} L^{a\alpha}_r \overline{H}_a (e^c)^\beta_s. \tag{2}$$

The superscript c denotes charge conjugation. These are all-left-Weyl expressions, not four-component barred expressions with unchanged charges.

Theorem A.1 (Conditional local action composition)**.** *Under the stated representation, Spin and kinetic assumptions, the density*

$$\begin{aligned}\mathcal{L} = {} & \frac{1}{2} \sum_{s=3,2,1} \kappa_s \operatorname{Tr}(F^s_{\mu\nu} F_s^{\mu\nu}) + \sum_{r,\psi} \frac{i}{2} \left[\psi^\dagger_r \bar{\sigma}^\mu D_\mu \psi_r - (D_\mu \psi_r)^\dagger \bar{\sigma}^\mu \psi_r \right] \\ & + (D_\mu H)^\dagger D^\mu H - m^2 H^\dagger H - \lambda (H^\dagger H)^2 - \sum_{r,s,\chi=u,d,e} \left(Y^\chi_{rs} T^\chi_{rs} + \overline{Y^\chi_{rs}} (T^\chi_{rs})^\dagger \right)\end{aligned} \tag{3}$$

is Grassmann even, Hermitian and invariant under arbitrary infinitesimal local gauge transformations. The real m^2, λ, positive κ_s and complex Y^χ_{rs} remain free coefficients. The construction composes varying-field gauge, scalar and fermion kinetic terms and all three Yukawa channels.

Proof. For an anti-Hermitian local parameter θ, set $\delta\psi = \rho(\theta)\psi$, $\delta H = \rho_H(\theta) H$ and $\delta A_\mu = -\partial_\mu \theta + [\theta, A_\mu]$. The product rule gives

$$\delta(D_\mu \psi) = \rho(\theta) D_\mu \psi, \qquad \delta F_{\mu\nu} = [\theta, F_{\mu\nu}].$$

The first identity cancels the two opposite $(\partial_\mu \theta)\psi$ terms. The second uses symmetry of the second parameter jet and cancels the first-derivative terms against the connection commutators. No equation of motion is used. Cyclic trace proves gauge-kinetic invariance; anti-Hermiticity and commutation of internal generators with $\bar{\sigma}^\mu$ prove kinetic invariance. The trace forms are ad-invariant, with the free normalizations permitted by the invariant-form classification.

The color fundamental–dual pairing and weak determinant tensor are invariant. The hypercharge sums are $1/6 + 1/2 - 2/3 = 0$, $1/6 - 1/2 + 1/3 = 0$ and $-1/2 - 1/2 + 1 = 0$. The supplied Spin epsilon contracts anticommuting Weyl components. Each T^χ_{rs} and its adjoint are gauge neutral, separately for every family pair. Every term has even Grassmann degree and the displayed conjugate combinations are Hermitian. The symmetrized kinetic expression differs from $i\psi^\dagger \bar{\sigma}^\mu D_\mu \psi$ by a total derivative. □

Proposition A.2 (Matter current of the same local action)**.** *For an anti-Hermitian basis element T_a, define $\delta\mathcal{L}_{\text{matter}} = j^\mu_a \delta A^a_\mu$ with matter jets fixed. Then*

$$j^\mu_a = i \sum_{r,\psi} \psi^\dagger_r \bar{\sigma}^\mu \rho_\psi(T_a) \psi_r + (\rho_H(T_a) H)^\dagger D^\mu H + (D^\mu H)^\dagger \rho_H(T_a) H. \tag{4}$$

Its covariant divergence vanishes on the matter Euler–Lagrange equations, using the dual adjoint connection on the current index.

Proof. Differentiate the two covariant kinetic terms. Anti-Hermiticity doubles the fermion contribution in the symmetrized expression; the potential and Yukawa tensors have no connection dependence. For a compactly supported gauge variation, the matter equations remove matter variations from the first variation of the action. Substituting $\delta A_\mu = -\partial_\mu \theta + [\theta, A_\mu]$ and integrating by parts gives the dual covariant-divergence identity. □

Exact local checks and scope. The executable fixture retains three families, 900 independent Grassmann value, derivative and conjugate generators, and all 27 complex Yukawa slots. Connection and scalar jets are exact Gaussian-rational test values, not physical coupling constants or measurements. Independent arithmetic checks reconstruct the density, retain first and symmetric second jets of the gauge parameter, and reject omitted inhomogeneous connection terms, omitted curvature commutators, wrong dual color action and mixed Yukawa charges. Grassmann anticommutation is checked separately from neutrality. Lean proves the noncommutative first- and second-jet identities, the cyclic-trace Ward identity and the spin-sign control. The full action and on-shell current argument are analytic, not full Lean formalizations.

The flat chart, Spin interpretation, kinetic choice, Higgs/Yukawa content, relative normalizations, family multiplicity and local connection remain declared. Identifying the connection with one reconstructed from source response histories is separate. There is no source-produced Standard Model action or current, broken-vacuum selection, gauge fixing or ghost construction, quantum anomaly cancellation from a fermionic measure, renormalization, continuum quantum-field existence theorem or numerical particle prediction. Global gauge quotient and physical line-spectrum attachments remain separate.

A.2 Finite current realizations and their domains

The local action above fixes which charge and current must be transported into an executable realization. For a number-conserving CAR hopping term

$$\widehat{H}_{xy} = c_x^\dagger U_{xy} c_y + c_y^\dagger U_{xy}^\dagger c_x,$$

its link derivative is the current of that same Hamiltonian. With a consistent orientation, $\dot{Q}_x = i[\widehat{H}, Q_x]$ is the signed sum of these link currents. This identity survives exact finite Fock construction; interpreting a coefficient matrix as a measured charge requires the same state, generator, clock and readout on both sides.

The source implementations in `code/sm_fermion_current/` retain the registered representations and all outcomes of their declared operations. Their hybrid branch has classical compact links and quantum CAR matter, not a fully quantized nonabelian gauge field. Internal scalar covariances give zero expectation of each traceless nonabelian source. The corresponding quantum charge variance can remain positive, so this is not a sharp nonabelian Gauss-law state. Its specified electric kinetic function satisfies

$$T'_\lambda(E) = \frac{2}{\lambda} \arctan \frac{\lambda E}{2M}, \qquad 0 \le \frac{E^2}{2M} - T_\lambda(E) \le \frac{\lambda^2 E^4}{48 M^3}.$$

This is a controlled comparison of energy functions, not a trajectory or continuum theorem. Cartan scalar reductions likewise retain both gauge variations of the action rather than deleting a force by setting its field to zero.

The particle construction gives a stronger same-operator finite attachment: the actual twelve current matrices act on an isolated spatial band tensored with the exterior matter module and the declared family space [1]. For the radial zero-mode band, the localized CAR map intertwines those currents and the specified P-dependent pairing operator. A compact defect pair instead has a four-dimensional band with nonzero spatial matrix A, and its one-body energies are shifted by $\lambda(A)$. A finite quartic interaction has an exact radial CAR restriction, but it changes charged poles and screening response. None of these identities selects a continuum chiral field, removes spatial doublers or derives a physical spin-statistics attachment from the gauge algebra alone.

B Scope of the gauge conclusion

The complete-response and inner-transport premises determine the compact Lie type. The declared matrix embedding, exterior module and band rule give a conditional one-generation representation and family architecture. The local action and finite current identities join those algebraic data to explicit operators. A physical realization must additionally identify the same local gauge field, constrained state space and renormalized current on one interacting continuum limit. Coupling selection, the vacuum, defect species, pole masses and gravitational response require that common model; they are not fixed by dimension twelve or by the Lie-type theorem.

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